**1. 8-Puzzle Problem**

**Aim:**

To write a Python program to solve the 8-Puzzle problem using a search algorithm.

**Algorithm:**

1. Start the program.
2. Define the initial state and goal state.
3. Represent the puzzle using a 3 × 3 matrix.
4. Find the empty space (0).
5. Generate possible moves by moving the empty space up, down, left, or right.
6. Compare the current state with the goal state.
7. Repeat until the goal state is reached.
8. Display the solution path.
9. Stop.

**Code:**

```
start = (1, 2, 3,
         4, 0, 6,
         7, 5, 8)

goal = (1, 2, 3,
        4, 5, 6,
        7, 8, 0)

def solve(start):
    q = deque([(start, [])])
    visited = {start}

    while q:
        state, path = q.popleft()

        if state == goal:
            return path + [state]

        zero = state.index(0)
        r, c = divmod(zero, 3)

        for dr, dc in [(1,0), (-1,0),
                       (0,1), (0,-1)]:
            nr, nc = r + dr, c + dc

            if 0 <= nr < 3 and 0 <= nc < 3:
                pos = nr * 3 + nc
                new = list(state)
                new[zero], new[pos] = new[pos], new[zero]
                new = tuple(new)

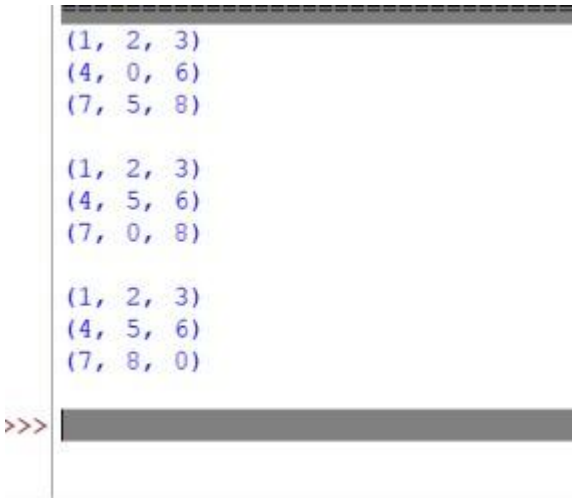
                if new not in visited:
                    visited.add(new)
                    q.append((new, path + [state]))

    return None

result = solve(start)

if result:
    for state in result:
        print(state[:3])
        print(state[3:6])
        print(state[6:])
        print()
```

**Output:**




**Result:**

Thus, the Python program to solve the 8-Puzzle problem was executed successfully.